

Erdős Problem #809

EP-809

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and define $F_k(n)$ to be the minimal r such that there is a graph G on n vertices with $\lfloor n^2/4 \rfloor + 1$ many edges such that the edges can be r -coloured so that every subgraph isomorphic to C_{2k+1} has no colour repeating on the edges.

Is it true that $F_k(n) \sim n^2/8$?